WISMITHA PEIRIS

Electronic & Embedded Engineer | PCB Design • Embedded ML • Robotics & AI

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PROFESSIONAL SUMMARY

Electronic Engineer with end-to-end experience across PCB design, sensors, embedded machine learning, mobile applications, and cloud pipelines. Recent work spans an industrial predictive-maintenance platform combining a custom power-switching PCB, sensor acquisition, ML models, and AWS storage; a children-interaction robot prototype using real-time LLM, STT, and TTS pipelines with computer vision and a projector-based interactive UI; and a published Android event-management product live at eventplanners.app. BSc Hons in Electrical and Electronic Engineering from the University of Peradeniya, with an MSc in Electronic and Electrical Engineering currently in progress at Anglia Ruskin University, UK. Comfortable shipping working systems under real-world constraints, from hardware bring-up through software integration.

TECHNICAL SKILLS

Programming Python (primary), C, C++, Kotlin, ESP-IDF, Bash
Hardware & PCB PCB design (Altium Designer, KiCad, EasyEDA), schematic capture, hardware bring-up, failure analysis, functional testing, JTAG, oscilloscope/DMM, soldering, electrical control panel design
Embedded & Microcontrollers Raspberry Pi (Pi 4 / Pi 5), ESP32, Arduino (Mega 2560), Jetson AGX; ADC, I2C, SPI, UART, Modbus, real-time multi-threaded acquisition
Sensors & DAQ Industrial vibration, current, and temperature sensors; high-speed real-time data acquisition; sensor calibration, synchronization, signal processing
ML & AI Time-series anomaly and fault detection, model optimization, edge / on-device deployment, LLM + STT + TTS pipelines, computer vision, OCR, automated WER/CER evaluation
Robotics & Controls Model Predictive Control (MPC), LiDAR mapping, path planning, autonomous navigation
Cloud & Data AWS (S3, data pipelines, Parquet / columnar storage and compression), Firebase (Firestore, Auth, Storage, Cloud Functions), Streamlit
Mobile (Android) Kotlin, Jetpack Compose, MVVM, Google Play Billing, Chromecast integration
Industrial Automation PLC programming (Siemens, Delta), HMI design, pneumatic integration, transformer fundamentals (amorphous core), battery backup with high-speed power switching
Design & Simulation SolidWorks, SolidWorks Electrical, AutoCAD, Proteus

PROFESSIONAL EXPERIENCE

Prototype Engineer — Figmentum Robotics LLC (USA, Remote) Sep 2025 – Present
End-to-end prototype development for a children-interaction robot — embedded AI, vision, audio, and projector UI.

- Built ultra-low-latency LLM + STT + TTS conversational pipeline in Python; optimized end-to-end response time and streaming latency for real-time interaction with children.
- Developed vision-based educational games (handwriting recognition for spelling, written math problem solving with step-by-step LLM tutoring) combining computer vision and language model reasoning.
- Engineered projector-based interactive display with keystone correction and reliable optical alignment; integrated wake-word, dialogue, game sessions, and display refresh into a single embedded loop.
- Built and currently beta-test two production Python platforms — an audio simulation / transcription accuracy testing platform (multi-provider STT, 28+ audio transforms, conversation simulation, automated WER/CER reporting) and an OCR simulation platform (synthetic image generation, model benchmarking, accuracy/latency measurement).
- Evaluated edge controllers and accelerators (latency, thermals, power) to select target hardware for on-device LLM and vision deployment.

Electronic Project Engineer — Hypen Holdings (Pvt) Ltd Nov 2024 – Aug 2025
Horana, Sri Lanka. Owned full predictive-maintenance hardware platform: sensors → edge → ML → cloud.

- Programmed industrial vibration, current, and temperature sensors for high-speed real-time data acquisition on Raspberry Pi 4 (Python); designed sampling and synchronization for harsh industrial conditions.
- Designed overall system architecture spanning sensor interfacing, edge processing, ML inference, and cloud storage.
- **Designed PCB (schematic to layout)** for a custom battery-backup module with high-speed power switching — guarantees uninterrupted operation through line-power loss in harsh environments.
- Trained and optimized ML anomaly / fault-detection models on time-series sensor data; iteratively tuned features and architecture for higher accuracy and lower inference latency.
- Implemented AWS data storage pipelines using Parquet and other columnar / compression formats to reduce storage cost and accelerate retrieval for predictive-maintenance analytics.

Trainee Robotics & Automation Engineer — Midas Safety (Pvt) Ltd Jul 2023 – Oct 2023
Biyagama & Awissawella, Sri Lanka. Industrial automation and PLC programming on glove-manufacturing production lines.

- Designed and built a production heat-transfer machine for glove printing: Delta PLC programming, HMI UI design, electrical panel design, hardware wiring, and pneumatic integration; deployed on the Awissawella floor.
- Designed a secondary electrical control panel synchronized with the main panel that measurably reduced glove defect rate by improving process control.
- Conducted plant-wide electrical panel survey identifying non-compliant and unsafe electrical conditions across the facility.

Trainee Electrical Engineer — LTL Transformers (Pvt) Ltd May 2022 – Aug 2022

Moratuwa, Sri Lanka. R&D in distribution-transformer materials science.

- Conducted amorphous-core transformer R&D project, evaluating core materials to reduce energy losses in distribution transformers.

PROJECTS

Autonomous Vehicle — Vision, LiDAR & Real-Time Dashboard • *Independent Engineering Project* 2025 – 2026
End-to-end software platform on Raspberry Pi 5 with dual IMX500 AI cameras (front-facing for lane and obstacle detection, rear/bird's-eye for smooth driving), TF Luna LiDAR for distance-based braking, MG996 servo steering, and L298N motor driver. Built a finite state machine across manual, autonomous, and calibration modes; computer-vision lane detection with ROI and binary thresholding; PID steering with anti-windup; and a real-time web dashboard featuring dual camera feeds, telemetry, live calibration, latency monitoring, and emergency stop.

Children-Interaction Robot — Real-Time Multi-language AI • *Figmentum Robotics LLC* 2025 – 2026
Built ultra-low-latency LLM + STT + TTS conversational pipeline; vision-based spelling and math educational games; projector-driven interactive display with keystone correction; researched on-device deployment across edge controllers and accelerators.

Audio Simulation & Transcription Accuracy Testing Platform • *Figmentum Robotics LLC* 2025 – 2026
Streamlit/Python platform with multi-provider STT (GPT-4o, Gemini, Whisper, Deepgram, Azure, AssemblyAI), 28+ audio transform blocks, conversation simulation with VAD/AEC/barge-in detection, and automated WER/CER, keyword accuracy, diarization, and prosody evaluation. Currently in beta.

OCR Simulation & Testing Platform • *Figmentum Robotics LLC* 2025 – 2026
Synthetic handwriting image generation (controllable style, paper, lighting, geometric distortion), model benchmarking framework, resolution / lighting / perspective-warp simulation, and labeling workflow for fine-tuning. Covers words, math equations, and paragraphs. Currently in beta.

Event Planner — Digital Invitations & Event Management Android App • *WellSmart Automations (live: eventplanners.app)* 2025 – 2026
Independently designed, developed, and deployed full-featured Android app (Kotlin, Jetpack Compose, Firebase, MVVM): customizable invitations with 18+ fonts and template editor, dynamic single-link RSVP/gallery web pages, real-time RSVP tracking, QR check-in, drag-and-drop seating chart, budget tracker with PDF reports, photo gallery with Chromecast slideshow, AI-powered text generation, co-host real-time chat, and tiered subscription model with Google Play Billing.

Predictive Maintenance Hardware Platform • *Hypen Holdings (Pvt) Ltd* 2024 – 2025
Industrial Pi 4 platform integrating vibration, temperature, and current/power sensors; designed PCB for high-speed power switching in custom battery-backup module; built ML fault-detection models on time-series data; deployed AWS pipeline using Parquet for cost-efficient storage and retrieval.

Object Avoidance Robot using Model Predictive Control (MPC) • *University of Peradeniya* 2024
Designed mechanical structure (CAD), integrated LiDAR for real-time mapping and obstacle detection, implemented MPC for path planning. Full onboard programming on Jetson AGX delivered reliable autonomous navigation.

Heat Transfer Machine — Glove Printing Automation • *Midas Safety (Avissawella)* 2023
Programmed Delta PLC, designed HMI UI, integrated pneumatics with the electrical panel for a production-deployed glove-printing automation system.

Smart Home Automation System (ESP32) • *University of Peradeniya* 2023
Designed PCB in KiCad and firmware in ESP-IDF for an ESP32-based smart-home controller integrated with Google Home and Alexa, with a touchscreen for local control and settings management.

Double-Drive Design for Optimizing Glove Manufacturing • *Midas Safety (Avissawella)* 2023
Designed and integrated a new electrical control panel synchronized with the main panel — reduced glove defect rate by enhancing process control and overall production efficiency.

Vending Machine Prototype • *University of Peradeniya* 2022
Built a functional prototype of a modern vending machine using CAD and Arduino Mega2560-based control systems, replicating commercial vending-machine functionality (coin acceptance, item selection, dispensing, status display).

EDUCATION, MEMBERSHIP & ACTIVITIES

MSc, Electronic and Electrical Engineering — Anglia Ruskin University, UK 2025 – 2026
BSc Eng. (Hons), Electrical and Electronic Engineering — University of Peradeniya, Sri Lanka 2020 – 2024
Programming for Everybody (Python) — University of Michigan (online) 2020
C for Everyone: Programming Fundamentals — University of California (online) 2020
Diploma in Western Music — Trinity College London Examinations Completed 2018
Associate Member, Institute of Engineers Sri Lanka (IESL) | Member, University of Peradeniya Athletics Team (2020 – 2024)